

Genesis Observatory Specification (GOS)

Project Genesis Research Repository

Generated: July 2026

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Version 1.0 July 2026

The Observatory (Evidence Explorer) is the scientific instrument deck of the Genesis Portal. Its role is to explain the narrative: if the Civilization Record answers \textit{what} happened, the Observatory answers \textit{why} it happened using quantitative evidence.

1. Core Philosophy

In accordance with \href{file:///c:/Users/vinay/Desktop/project%20genesis/docs/architecture/ADR/ADR-006-narrative-first.md}{ADR-006} and \href{file:///c:/Users/vinay/Desktop/project%20genesis/docs/architecture/ADR/ADR-007-goal-oriented-visualizations.md}{ADR-007}, the Observatory operates under two absolute constraints:

1. \textbf{Evidence supports narrative}: The Observatory is entered \textit{from} the Civilization Record. It is a secondary explorer, not a standalone home page.
 2. \textbf{Visualizations are Laboratory Instruments}: Every chart must answer a specific research question. Generic data plots without intent are prohibited.
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2. Instrument Catalog & Registry

We document and register all approved scientific instruments. Every instrument must map to a research question, inputs, and limitations.

OBS-001: Population Dynamics (Oscilloscope)

- \textbf{Research Question}: \textit{Did resource scarcity reduce survival and growth rates?}
- \textbf{Inputs}: \texttt{population.csv} (ticks, total count, and colony breakdowns).
- \textbf{Outputs}: Line graph mapping total population and individual colony lineages over time.
- \textbf{Limitations}: Does not show spatial distribution or local water pressures.

OBS-002: Resource Consumption Explorer (Flow Meter)

- \textbf{Research Question}: \textit{Where were critical resource shortages created?}
- \textbf{Inputs}: \texttt{summary.json} (derived metrics for food and water efficiency).
- \textbf{Outputs}: Stacked area chart showing resource supply vs. total agent consumption.
- \textbf{Limitations}: Aggregated globally; does not show individual agent hunger levels.

OBS-003: Territorial Friction Map (Microscope)

- \textbf{Research Question}: \textit{Did territorial pressure lead to clustered disputes?}
- \textbf{Inputs}: \texttt{events.json} (Dispute events metadata mapping coordinates).
- \textbf{Outputs}: Static 2D spatial heatmap overlay showing conflict coordinates on the map.
- \textbf{Limitations}: Ticks are collapsed; does not show the sequence of dispute progression.

OBS-004: Social Kinship Grid (Oscilloscope)

- \textbf{Research Question}: \textit{Did stable cooperative attachment bonds emerge?}
- \textbf{Inputs}: \texttt{agent_census.csv} (parents, colonies, and children lineages).
- \textbf{Outputs}: Dynamic directed network graph showing colony splits and child-parent attachments.
- \textbf{Limitations}: Becomes visually complex when agent count exceeds 500.

OBS-005: Genetic Entropy Browser (Sequencer)

- \textbf{Research Question}: \textit{Did genetic traits stabilize or drift over generations?}
 - \textbf{Inputs}: \texttt{agent_census.csv} (chromosome strings and generation indices).
 - \textbf{Outputs}: Histogram showing distribution of mutation deviations from the founder seed.
 - \textbf{Limitations}: Captures survivors only; deceased lineages are excluded.
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3. Interaction & Linking Rules

- \textbf{Tick Linking (Phase 3.5 Bridge)}: Clicking on any event in the Chronicle seeks all active instruments to that specific tick (e.g. updating the pointer on the Population Dynamics graph).
- \textbf{Decoupled Tabs}: The Observatory loads in a split-screen or separate tab context, keeping the narrative chronicle layout readable.